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Averaged Bi-LSTM networks for RUL prognostics with non-life-cycle labeled dataset

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ABSTRACT

LSTM network is an effective RNN model to predict the system RUL for its superior performance in sequential data processing. Usually, networks trained by life-cycle labeled dataset would possess similar RUL predicting accuracies, because the network training algorithm could ensure the network optimality for the whole training dataset. However, for networks trained by non-life-cycle labeled samples, the network uncertainty caused by different training conditions could lead to degradation prediction uncertainty for some local points. Further, the RUL predicting results that are computed by these uncertain local points may shows relatively large differences. Therefore, in order to obtain an accurate RUL prediction with networks trained by non-life-cycle labeled samples, our paper proposes a novel network model averaging method to deal with the network uncertainty. What is more, to learn the temporal correlation information of training samples sufficiently, we adopt the Bi-LSTM network to illustrate the application of the proposed network model averaging method. Finally, degradation values of Graphite/LiCoO2 battery are used to verify the effectiveness of the proposed method. The results show that the proposed method could improve the RUL prediction accuracy and reduce the prediction error.

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1. Introduction

A single failure of complex-huge systems in manufacture, aerospace engineering and transportation can lead to catastrophic consequences with profound impacts and potential safety hazards. Therefore, remaining useful life (RUL) prognostics has been termed as the important method to ensure the operation safety [1–3]. Based on current research achievements in RUL prognostics, model-based prognostics and data-driven approaches are two main broad categories [4]. Model-based prognostics are effective methods to predict the system degradation, forexample crack growth model [5] which has been widely adopted in prognostics of aircraft, industrial and manufacturing processes. However, because the degradation mechanism of practical system is complex, it is hard to build an accurate physics degradation model. Therefore, data-driven approaches have become more and more popular in system RUL predicting.

Data-driven approaches mainly include support vector machine(SVM) [6], relevance vector machine (RVM) [7], Hidden Markov Model (HMM) [8] and neural network (NN) method [9]. These methods highly depend on the degradation data to acquire the predicting results. With the development of sensor technology and computer science, huge amount of degradation data have been observed and available. NN method has become an important subcategory of data-driven prognostics [10]. Deutsch et al. [11] adopted Restricted Boltzman Machine (RBM) to predict the RUL of bearing. This kind of NN could extract important features that can be used for RUL estimation automatically. Babu et al. [12] and Li et al. [13] used the deep Convolutional Neural Network (CNN) when predicting the RUL of aero-engine. In order to incorporate automated feature learning, they applied the convolution and pooling filters along the emporal dimension over the multichannel sensor data. Although RBM and CNN perform well in RUL predicting, they could not learn the temporal correlation information of training dataset sufficiently compared with Recurrent Neural Network (RNN) [14–16]. Because the inner cell of RNN contains recurrent connections, RNN could transmit the previous information to current cell state and deal with the temporal correlation information well. Heimes [17] and Gugulothu et al. [18] adopted RNN to obtain the RUL of aero-engine, and Liu et al. [19] used RNN to estimate the degradation of battery. These researches show that RNNs are good at processing sequential data. However, RNNs

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that consist of standard recurrent cells are not capable of handling "long-term dependencies" [20]. Therefore, long short-term memory(LSTM) nets, which are proposed by Hochreiter et al. [21], have become the focus of RUL predicting researches. Yuan et al. [22], Zheng et al. [23] and Wu et al. [24] constructed LSTM networks to process the degradation of aero-engine, and the results show that LSTM nets outperforms the traditional approaches for RUL estimation. Zhang et al. [25] bulit an two-hidden-layer LSTM network to deal with the degradation of lithium-ion batteries. They optimized the LSTM network through resilient mean square backpropagation technique and dropout technique, and the predicted RULs of lithium-ion batteries show that LSTM net outperforms the SVM and original RNN. Besides the selection of network cells, the network structure optimization is also a way to improve the network performance. Huang et al. [26] used the bidirectional LSTM (Bi-LSTM) network to predict the RUL of turbofan engine. Because the Bi-LSTM network could learn the temporal correlation informations in both forward and backward time directions simultaneously, this network could obtain a higher prediction accuracy compared with the stacked LSTM networks. Therefore, in order to handle the problem of "long-term dependencies" and learn the temporal correlation information of training samples sufficiently, the selection of network architecture is also important.

Network training is the first step to utilize the constructed nets. Usually, for a given training dataset and a specific network framework, the obtained networks' weights are slightly different from each other, when the training hyper-parameters settings such as batch size, learning rate and epoch times, are various. Under different training conditions, the RUL predicting networks trained by life-cycle labeled dataset would possess similar RUL predicting accuracies, because the network training algorithm could ensure the network optimality for the given training dataset. However, when the constructed networks are trained by non-life-cycle samples forexample degradation value labeled data, these networks would have approximate global degradation prediction accuracies based on the well-behaved training algorithm, but may show a relatively large gap at some local predicted degradation points. This is because that the training algorithm only focuses on the network optimization about the whole training dataset and does not care about local degradation points. Therefore, the RUL predicting results that are further computed by these local predicted degradation points may shows relatively large differences. This means that the network uncertainty caused by training conditions could lead to prediction uncertainty for some local prediction points. Further, the prediction uncertainty of degradation value would lead to RUL prediction uncertainty. Considering that non-life-cycle samples are widely used in current papers such as bearing datasets [27–30], lithium-ion battery datasets [31–34] and the metal crack propagation dataset [35], network model uncertainty is a question worth researching [36].

Bayesian model averaging is an effective method to deal with the model uncertainty [37–40], but the use of integrated likelihood function makes it impossible to be adopted in network models. The mathematical expressions of network models are complicated, and their likelihood functions are usually hard to be obtained. Thus, combining the advantages of Bayesian model averaging and features of network models, our paper proposes a novel network model averaging method to deal with uncertainty of networks trained by non-life-cycle labeled dataset.

The rest of the paper is organized as follows. Section 2 explains the Bi-LSTM network constructing and training, and introduces an optimized regularization method to prevent overfitting in network training. Section 3 analyzes the network model uncertainty and proposes our network model averaging method. Section 4 shows the results of experimental study. A short conclusion with an emphasis on future research challenges is given in Section 5.

2. LSTM network constructing and training

2.1. LSTM cells and Bi-LSTM network

LSTM nets are composed of LSTM cells in a specific network architecture. In order to make clear the LSTM network model, we first introduce the mathematical expressions of LSTM cells. RNNs consisting of sigma cells or tanh cells are unable to learn the relevant information of input data when the input gap is large. In order to deal with the problem of "long-term dependencies", Hochreiter et al. [21] proposed the LSTM cell. They improved the remembering capacity of the standard recurrent cell by introducing "gate" into the cell. Since this pioneering work, LSTM cells have been modified and popularized by many researchers [41,42]. Variations include LSTM without forget gate, LSTM with forget gate, and LSTM with peephole connections. Considering the fact that LSTM with forget gate is the most popular used, this kind of LSTM cell is adopted as the basic cell structure in our paper. Fig. 1 shows its inner connections.

Based on the connections shown in Fig. 1, the LSTM cell can be mathematically expressed as:

$$\begin{aligned} f_t &= \sigma(W_{fh}h_{t-1} + U_{fx}x_t + b_f) \\ i_t &= \sigma(W_{ih}h_{t-1} + U_{ix}x_t + b_i) \\ o_t &= \sigma(W_{oh}h_{t-1} + U_{ox}x_t + b_o) \\ \tilde{c}_t &= \tanh(W_{\tilde{c}h}h_{t-1} + U_{\tilde{c}x}x_t + b_{\tilde{c}}) \\ c_t &= f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \\ h_t &= o_t \cdot \tanh(c_t) \end{aligned} \quad (1)$$

where f_t, i_t and o_t denote the forget gate, input gate and output gate respectively, c_t is the cell state, h_t is the hidden state, x_t is the cell input, W and U are the weights of hidden state and cell input.

Due to the limited capacity of the single cell in handling engineering problems, LSTM cells have to be organized into a specific network architecture when processing practical data. Because the bidirectional LSTM (Bi-LSTM) network could learn the temporal correlation informations in both forward and backward time directions simultaneously, we adopt Bi-LSTM network to estimate the RUL of battery degradation. This kind of architecture consists of forward layers and backward layers, and the internal connections of recurrent layers are shown in Fig. 2.

2.2. Overfitting and regularization

The untrained Bi-LSTM nets could not be used to predict the system RUL, and network training is the first step to utilize it. However, because the constructed Bi-LSTM nets may possess excessive freedoms, overfitting may exist. When overfitting happens in network training, models will work well for the training dataset, but may fail to generalize to new situations. Therefore, in order to prevent LSTM networks from overfitting, we need to adopt the regularization method to deal with it. Dropout technique is an effective regularization method to prevent network overfitting. However, current dropout method is suitable for networks without recurrent connections [43,44], such as feedforward NN and CNN. Therefore, a novel dropout technique in depth dimension would be adopted in our paper.

Here, we first introduce the conventional dropout method. Current dropout method randomly and temporarily deletes some hidden layer neurons of the constructed networks and leaves the input and output layer neurons untouched. Fig. 3 shows the conventional dropout technique adopted in a three-hidden-layer feedforward network.

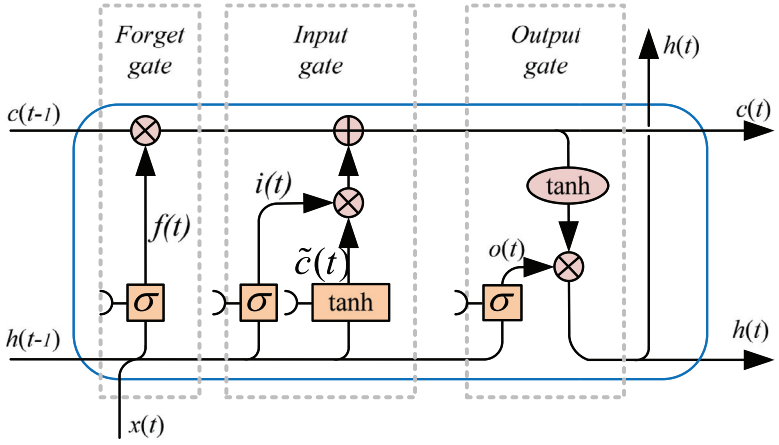


Fig. 1. The inner connections of LSTM with forget gate.

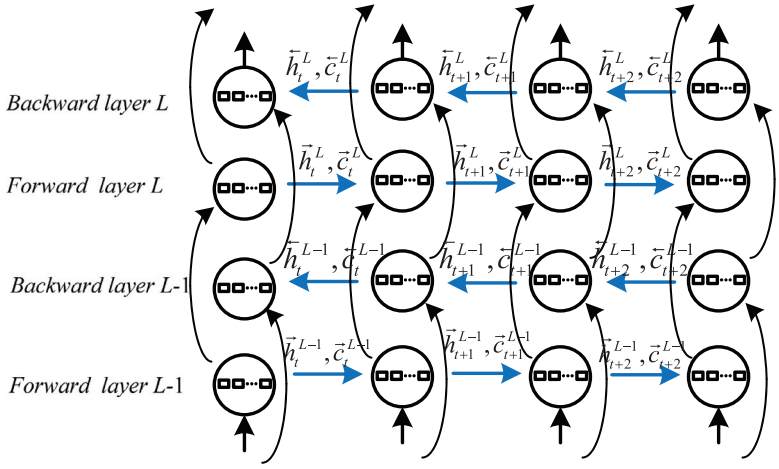
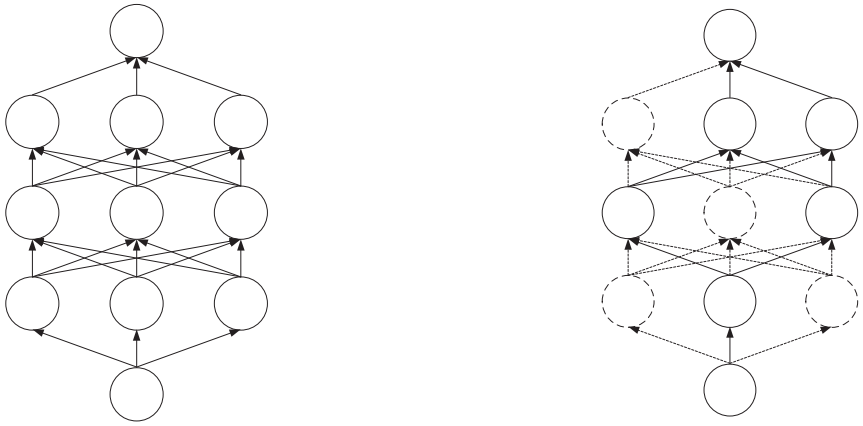


Fig. 2. Internal connections of Bi-LSTM network's recurrent layers.

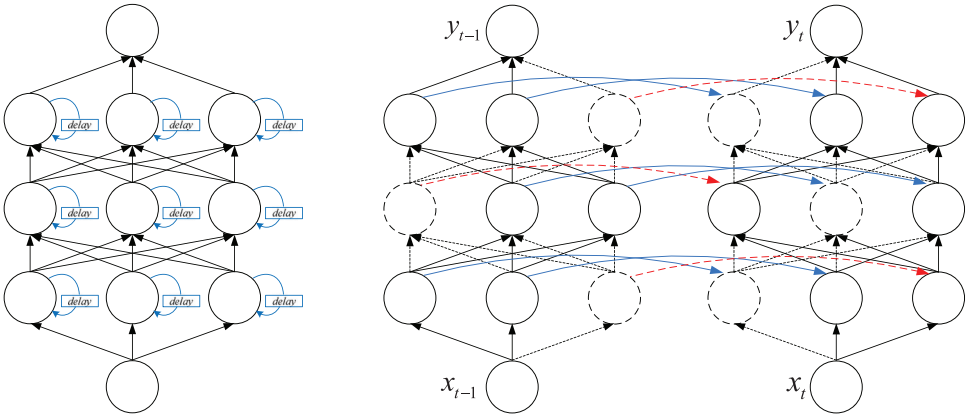


(a) The feedforward network without dropout technique (b) The feedforward network with dropout technique

Fig. 3. The conventional dropout technique adopted in a three-hidden-layer feedforward network.

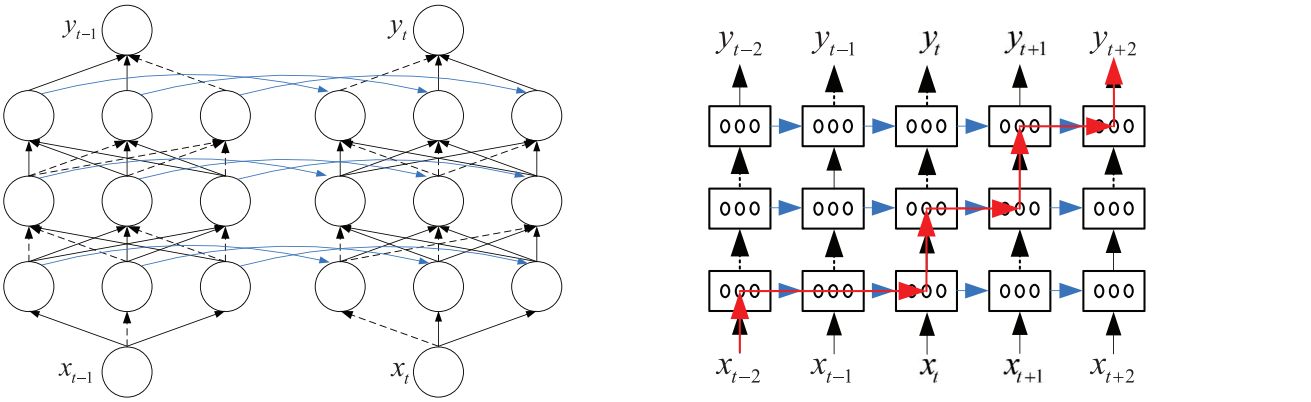
Through deleting some hidden layer neurons, conventional dropout technique could prevent networks from overfitting effectively. It has been proved that it is the most powerful regularization method for feedforward networks [45]. However, when adopted in training RNNs, the conventional dropout technique does not work well. Because it would perturb the recurrent connections of RNNs, which makes it difficult for RNNs to learn and store information for long periods of time. Fig. 4 explains this disadvantage with a three-hidden-layer RNN.

In Fig. 4, the black solid arrows indicate connections in depth dimension, the blue solid arrows indicate recurrent connections in time dimension, and the red dashed arrows indicate perturbed recurrent connections by conventional dropout techniques. In order to overcome this disadvantage, we should only apply dropout



(a) The RNN without dropout technique (b) The RNN with dropout technique

Fig. 4. The conventional dropout technique adopted in a three-hidden-layer RNN.



(a) The RNN with dropout in depth dimension (b) The typical information flow path of dropout in depth dimension

Fig. 5. Dropout in depth dimension technique.

method in depth dimension [46]. Fig. 5 shows the dropout technique in depth dimension.

Based on the inner connections in Fig. 5(a), we could formulate the LSTM cell at lth hidden layer in the network with dropout in depth dimension technique as:

$$\begin{aligned} f_t^L &= \sigma(W_{fh}^L h_{t-1}^L + U_{fx}^L D(h_t^{L-1}) + b_f^L) \\ i_t^L &= \sigma(W_{ih}^L h_{t-1}^L + U_{ix}^L D(h_t^{L-1}) + b_i^L) \\ o_t^L &= \sigma(W_{oh}^L h_{t-1}^L + U_{ox}^L D(h_t^{L-1}) + b_o^L) \\ \tilde{c}_t^L &= \tanh(W_{\tilde{c}h}^L h_{t-1}^L + U_{\tilde{c}x}^L D(h_t^{L-1}) + b_{\tilde{c}}^L) \\ c_t^L &= f_t^L \cdot c_{t-1}^L + i_t^L \cdot \tilde{c}_t^L \\ h_t^L &= o_t^L \cdot \tanh(c_t^L) \end{aligned} \tag{2}$$

where $D(\cdot)$ is the dropout operator that sets a random subset of its argument to 0.

3. RUL prediction algorithm and network averaging method

3.1. RUL prediction algorithm and uncertainty analysis

3.1.1. RUL prediction algorithm

In order to further explain our method, we first introduce the RUL prediction algorithm for the network model trained by degradation value labeled dataset. We express the trained RNN model M_c as:

$$y_{c,t+k} = f_c(x_{t-L+1}, x_{t-L+2}, \ldots, x_{t-1}, x_t), \tag{3}$$

where L and k are the length of input sequence and predicting interval steps respectively which are decided in network training process, x_t denotes the real degradation value at time t, $y_{c,t+k}$ is the predicted degradation value of model M_c. When adopting this network model to predict the degradation value, the first input should be $[x_1, x_2, ..., x_L]$, and $y_{c,L+k}$ would be the corresponding output. Let $RUL_{c,m}$ denote the predicted RUL of model M_c at the RUL prediction starting point $t = m$, and $X_m = [x_1, x_2, ..., x_m]$ denote the real degradation values from time $t = 1$ to m. Then the $RUL_{c,m}$ prediction algorithm could be expressed as:

In Fig. 6, when there is no real observed degradation value x_{m+w} at time $t = m + w$, we adopt $\bar{x}_{c,m+w}$, which equals to the predicted degradation value $y_{c,m+w}$ of model M_c at time $t = m + w - k$, as the network input. Therefore, we could use the predicted degradation values to predict the system RUL at any time not to be restricted by the predicting interval steps k.

3.1.2. Uncertainty analysis

Based on the above RUL prediction algorithm, some local predicted degradation values are adopted to compute the system RUL, when the target predicting point exceeds the predicting interval

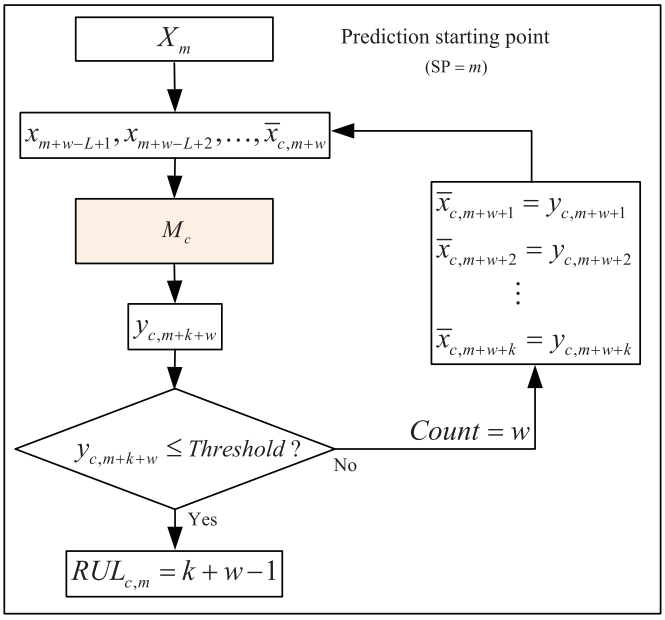


Fig. 6. The RUL prediction algorithm for network model trained by degradation value labeled dataset.

steps k. Therefore, it could be concluded that accurate predicted degradation values around RUL starting point $t = m$ are in demand, because these degradation values are the base of subsequent RUL predicting. However, when the training conditions are different, the obtained networks usually would have approximate global degradation prediction accuracies based on the well-behaved training algorithm, but may show a relatively large gap at some local predicted degradation points. This is because that the training algorithm only focuses on the network optimization about the whole training dataset and does not care about local degradation points. What is more, this means that the network uncertainty caused by training conditions could lead to prediction uncertainty for some local predicted degradation points. Further, the prediction uncertainty of the local predicted degradation values would lead to RUL prediction uncertainty. Thus, in order to obtain an accurate RUL prediction, network model uncertainty is a question worth considering

In order to verify our uncertainty analysis, we take some predicting results in Section 4 as examples in advance. Here, we adopt the Dataset-1 in Table 2 as the training dataset, and select the settings of M_0, M_2, M_4 and M_6 in Table 4 as the training hyper-parameters. Then, the obtained networks are used to predict the system RUL based on the RUL prediction algorithm in Fig. 6. Fig. 7 shows the diverse predicted trajectories of networks trained with different training hyper-parameters.

In Fig. 7, the predicted trajectories of different models are globally approximate and the maximal mean absolute error (MAE) between them is only 2.1×10^{-3}, but the maximal prediction error at some local prediction points nearly equals to 4.0×10^{-2}. Compared with the whole degradation amplitude, the local maximal prediction error is relatively small. However, these small degradation value errors would lead to large RUL uncertainty based on the RUL prediction algorithm in Fig. 6, because the predicted degradation values directly influence the subsequent RUL predicting. Fig. 8 shows the impacts of predicted degradation values on RUL predicting. In order to show the impacts of predicted degradation values on RUL predicting results clearly, the RUL predicting results of M_0 and M_2 at starting point SP = 300 are selected as the examples. What is more, according to the algorithm in Fig. 6, the number of the local degradation points used in RUL predicting process equals to the predicting interval steps k.

In Fig. 8(a) the MAE of the predicted degradation values which are used in RUL predicting is 3.4×10^{-3}, however, in Fig. 8(b) the obtained RUL error of M_0 and M_2 is 37, when the failure threshold is set as $Threshold = 0.5$. This means that a slight error of some local predicted degradation values could lead to the relatively large RUL predicting difference. In other words, the RUL prediction uncertainty is directly affected by prediction uncertainty of degradation value. Essentially, the prediction uncertainty comes from the network uncertainty caused by different training conditions. Therefore, in order to acquire a precise RUL prediction, the network model uncertainty should be considered.

3.2. Network model averaging method

3.2.1. Bayesian model averaging method and its restriction

Bayesian model averaging method is an effective method to deal with model uncertainty [37,47]. Let ∇ denote the expected utility, for example the RUL of target system. Then, considering Bayesian model averaging with the given data x, the posterior distribution of the expected utility can be expressed as:

$$P(\nabla|x) = \sum_{c=1}^{S} P(\nabla|M_c, x)P(M_c|x), \tag{4}$$

where $P(M_c|x)$ is the posterior probability of model M_c, and $P(\nabla|M_c, x)$ is the posterior distribution of the expected utility under model M_c. In the above equation, $P(M_c|x)$ is the weight of the model average, and could be calculated through:

$$P(M_c|x) = \frac{P(x|M_c)P(M_c)}{\sum_{i=1}^{S} P(x|M_i)P(M_i)}, \tag{5}$$

where $P(M_c)$ is the priori probability of model M_c, and $P(x|M_c)$ is the integrated likelihood function which means the probability of observing data x. We denote the parameter vector of model M_c by $\boldsymbol{\Theta}_c$ and parameter prior density by $P(\boldsymbol{\Theta}_c|M_c)$. Thus, the integrated likelihood $P(x|M_c)$ could be expressed as:

$$P(x|M_c) = \int L(x|\boldsymbol{\Theta}_c, M_c)P(\boldsymbol{\Theta}_c|M_c)d\boldsymbol{\Theta}_c, \tag{6}$$

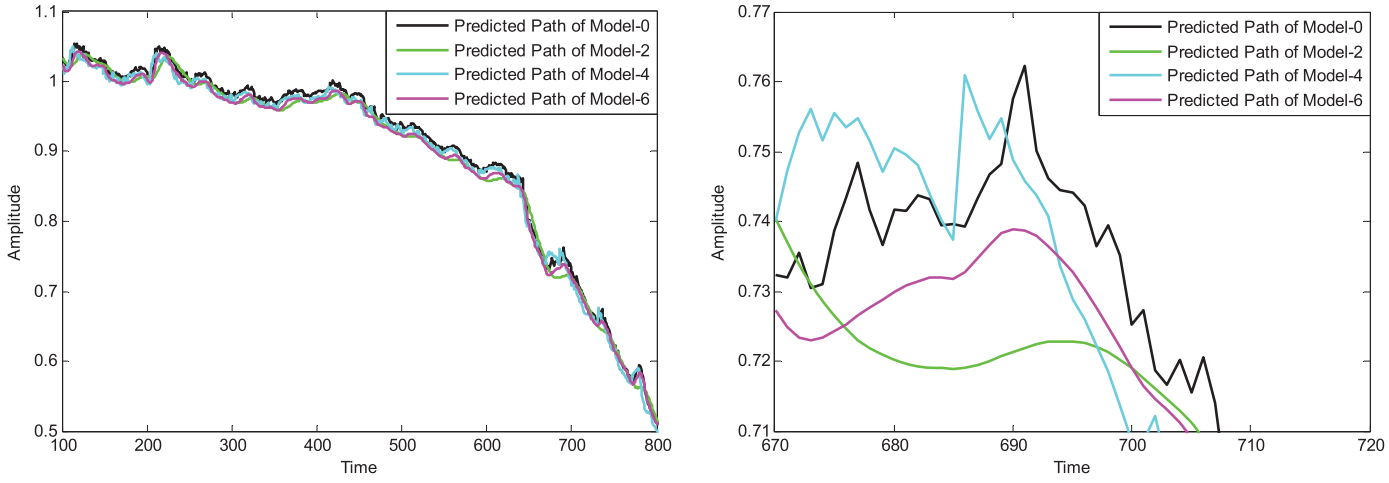
where $L(x|\boldsymbol{\Theta}_c, M_c)$ denote the likelihood function of model M_c.

In Eq. (9), the use of integrated likelihood function makes Bayesian model averaging method impossible to be adopted in the network model averaging. The mathematical expressions of network models are complicated, and the likelihood functions of networks are hard to obtain. Therefore, combining the advantages of Bayesian model averaging and the specialties of network's RUL prediction algorithm in Fig. 6, we propose a new network model averaging method to deal with the network model uncertainty.

3.2.2. Network model averaging method

In Fig. 6, because the RUL prediction algorithm is directly influenced by the k predicted degradation values which follow the RUL predicting starting point, the predicting accuracy of these local degradation value should be adopted to calculate the network probability, when dealing with the network uncertainty in RUL predicting. However, the real degradation value of these points could not be observed. Thus, combining the advantages of Bayesian model averaging and network models, we propose a novel network model averaging method.

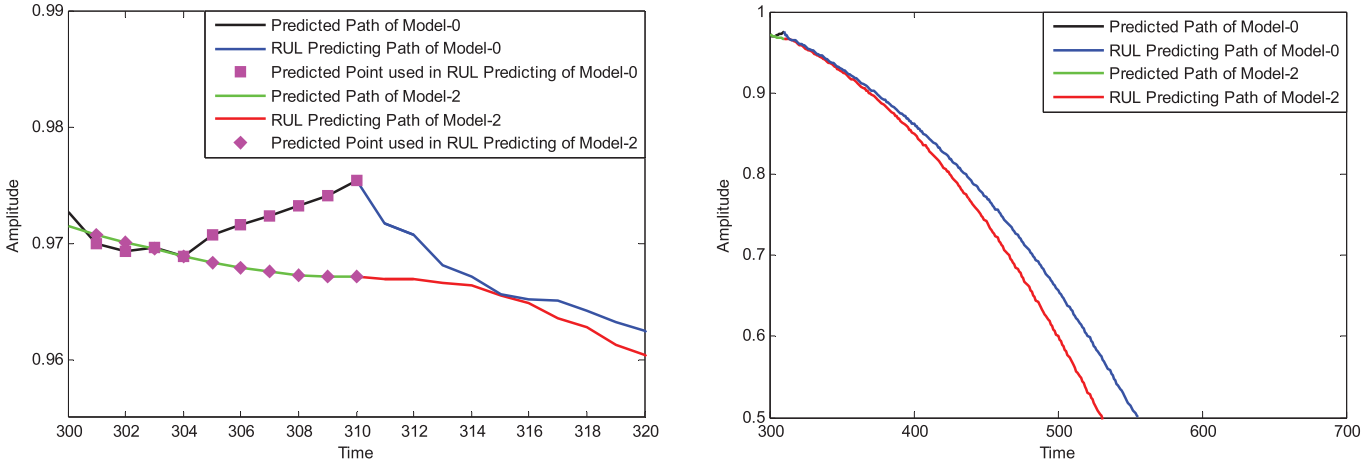
Considering the predicting datasets have some similar characteristics with the training dataset, networks with high global testing accuracy usually would work well on the predicting dataset. The global testing accuracy could be treat as the network prior probability just like in Bayesian model averaging. What is more, a



(a) The approximate global predicted trajectories

(b) The diverse local prediction values

Fig. 7. The predicted degradation trajectories of networks trained with different training hyper-parameters.



(a) The predicted degradation values used in RUL predicting

(b) The RUL predicting path based on predicted degradation values

Fig. 8. The results of RUL prediction algorithm.

small degradation prediction error often means that the predicted degradation values fit the real degradation well. Therefore, in order to reflect the relationship between the prediction error and accuracy, we define the global testing accuracy δ_c^{test} as:

$$\delta_c^{test} = 1 - \frac{1}{N}\sum_{i=1}^{N}\left[\sqrt{(x_i^{test} - y_{c,i}^{test})^2}/x_i^{test}\right], \tag{7}$$

where N is the number of testing samples. In Eq. (7), the testing degradation value x_i^{test} is positive, and the testing accuracy δ_c^{test} should always follow $\delta_c^{test} \leq 1$. A higher global testing accuracy means the trained networks would fit the predicting dataset better. Under this condition, the network should possess a larger prior probability. To ensure the practicality of our definition, we calculate the network prior probability by:

$$P_{TA}(M_c) = \begin{cases} \dfrac{\delta_c^{test}}{\sum_{i=1}^{S}\delta_i^{test}} = \dfrac{1 - \frac{1}{N}\sum_{i=1}^{N}\left[\sqrt{(x_i^{test} - y_{c,i}^{test})^2}/x_i^{test}\right]}{\sum_{j=1}^{S}\left\{1 - \frac{1}{N}\sum_{i=1}^{N}\left[\sqrt{(x_i^{test} - y_{j,i}^{test})^2}/x_i^{test}\right]\right\}}, & 1 \geq \delta_c^{test} \geq 0 \\ 0, & \delta_c^{test} < 0 \end{cases}. \tag{8}$$

Further, let $\delta_{c,m}^{pred}$ denote the global predicting accuracy of model M_c at starting point $t = m$. Then the global predicting accuracy $\delta_{c,m}^{pred}$ could be expressed as:

$$\delta_{c,m}^{pred} = 1 - \frac{1}{m-L-k+1}\sum_{j=L+k}^{m}\left[\sqrt{(x_j - y_{c,j})^2}/x_j\right]. \tag{9}$$

Similarly, the predicting accuracy $\delta_{c,m}^{pred}$ should always follow $\delta_{c,m}^{pred} \leq 1$. When the predicting accuracy $\delta_{c,m}^{pred}$ is higher, it means

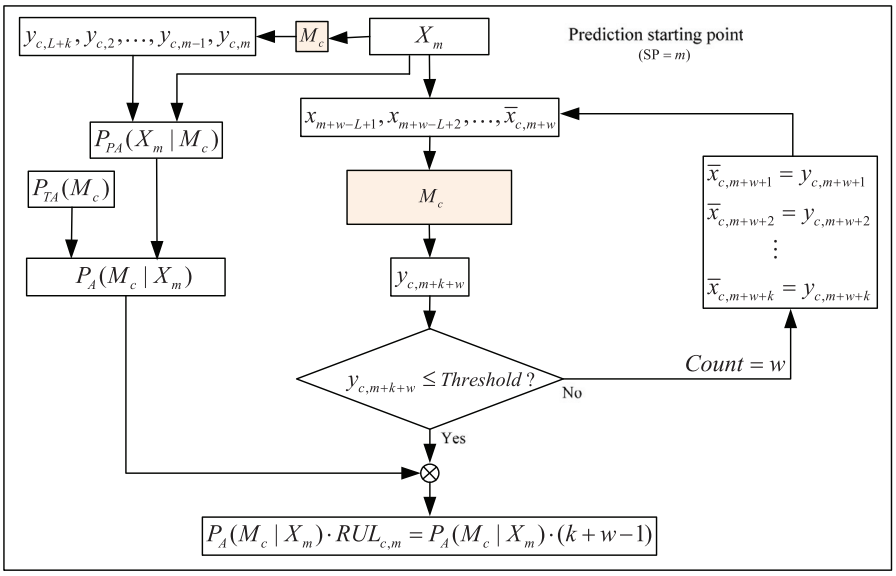


Fig. 9. The RUL prediction algorithm of model M_c with model posterior probability $P_A(M_c|X_m)$.

that model M_c could fit the predicting dataset better based on the observed real degradation values X_m from time $t = 1$ to m. Then, the probability of the observed real degradation values X_m should be higher, when M_c is the true network model for the predicting samples. Corresponding to the Bayesian model averaging method, we define the probability of X_m as:

$$P_{PA}(X_m|M_c) = \begin{cases} \dfrac{\delta_{c,m}^{pred}}{\sum_{i=1}^{S}\delta_{i,m}^{pred}} = \dfrac{1-\frac{1}{m-L-k+1}\sum_{j=L+k}^{m}\left[\sqrt{(x_j-y_{c,j})^2}/x_j\right]}{\sum_{i=1}^{S}\left\{1-\frac{1}{m-L-k+1}\sum_{j=L+k}^{m}\left[\sqrt{(x_j-y_{c,j})^2}/x_j\right]\right\}}, & 1 \ge \delta_{c,m}^{pred} \ge 0 \\ 0, & \delta_{c,m}^{pred} < 0 \end{cases} \tag{10}$$

Based on the network prior probability in Eq. (8) and the probability of X_m in (10), we could further define our novel model posterior probability as:

$$P_A(M_c|X_m) = \frac{P_{PA}(X_m|M_c)P_{TA}(M_c)}{\sum_{i=1}^{S} P_{PA}(X_m|M_i)P_{TA}(M_i)}. \tag{11}$$

The prediction accuracy $\delta_{c,m}^{pred}$ is updated each time a new degradation value x_m observed, so the defined model posterior probability $P_A(M_c|X_m)$ would also be updated in real-time. Based on the defined model posterior probability, we could calculate the averaged RUL_m through:

$$RUL_m = \sum_{c=1}^{S} P_A(M_c|X_m)RUL_{c,m}. \tag{12}$$

Fig. 9 shows the RUL prediction algorithm of model M_c with model posterior probability $P_A(M_c|X_m)$.

In Fig. 9, the detailed algorithm consists of the following steps:

- Step-1: Processing degradation data and constructing training dataset baed on the target problem analyzing.
- Step-2: constructing and training each candidate network model M_c.
- Step-3: Calculating the prior probability $P_{TA}(M_c)$ in Eq. (8) of network model M_c based on the training results of Step-2.
- Step-4: Calculating the network model posterior probability by Eq. (11).
- Step-5: Observing the RUL result of network model M_c based on the iterative predicting.
- Step-6: Calculating the system's RUL based on the model posterior probability $P_A(M_c|X_m)$ and RUL_m of each candidate model.

4. Experimental study

4.1. Data pre-processing

CS2 battery degradation datasets of CALCE Battery Research Group [48] are adopted in our reasearch, the details of the target CS2 battery are given in Table 1.

In the battery degradation test, four CS2 cells, CS2-35, CS2-36, CS2-37 and CS2-38, underwent the same charging profile which was a standard constant current/constant voltage protocol with a constant current rate of 1C until the voltage reached 4.2 V and then 4.2 V was sustained until the charging current dropped to below 0.05A. The discharge cut off voltage for these batteries was 2.7 V [49]. The discharge capacities of these four CS2 cells are selected as the degradation feature, and degradation trajectories are obtained by numerical integration method. Fig. 10 shows the discharge capacity values.

In order to obtain a well-performed network in the training process, the collected discharge capacity values are normalized to be within the range of [0,1]. Because the capacity rating of CS2 is 1100 mAh in Table 1, we think that the initial capacity of each standard CS2 cell should not exceed 1.21 Ah (110% capacity rating). Otherwise this CS2 cell should be identified as the inferior product and should not be used in the charging-discharging test. Therefore, the normalized data could be expressed as:

$$x_{normal}^{i,j} = x_{original}^{i,j}/1.21 \tag{13}$$

where $x_{original}^{i,j}$ is the i-th discharge capacity point of the j-th battery, $x_{normal}^{i,j}$ is the corresponding normalized value. In order to assess the performance of the proposed network model averaging method, and verify the Bi-LSTM network's evolvable properties and robustness to new situations, the observed discharge capacity values are adopted to build datasets in Table 2.

In Table 2, training samples CS2-35(0∼w%) are the obtained discharge capacity values of battery CS2-35 during time $0 < t \le T_{CS2-35} \cdot w/100$, where T_{CS2-35} is the total monitoring time of battery CS2-35, and predicting samples CS2-35(w%∼100%) are

Table 1
Graphite/LiCoO2 pouch cells.

CS2 Battery Sample	Battery Parameters	Specifications (Value)
	Capacity Rating	1.10Ah
	Cell Chemistry	LiCoO2 cathode
	Weight	21.1 g
	Dimensions (mm)	$5.4 \times 33.6 \times 50.6$
	Special Notes	Jellyroll configuration wrapped around the "length" axis, indicated by the red arrows below

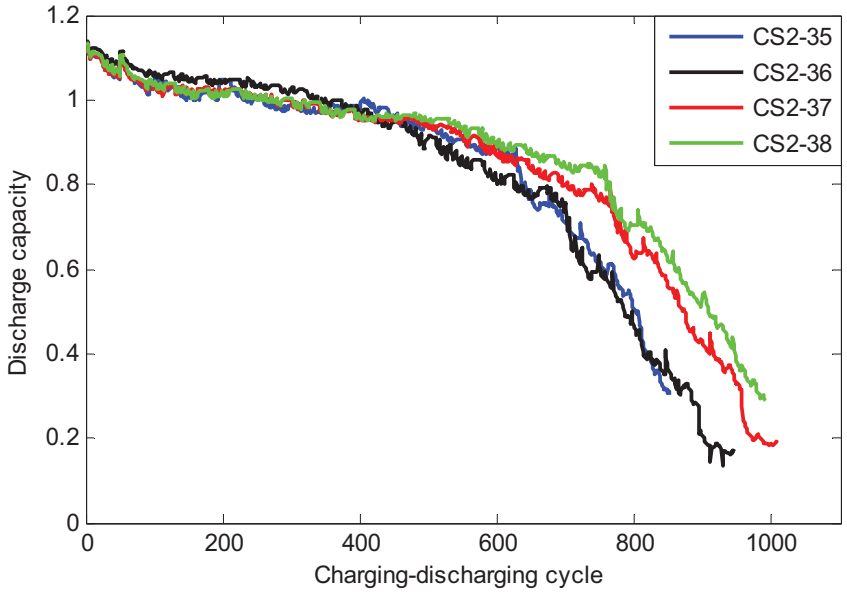


Fig. 10. The discharge capacity values.

Table 2
Details of the constructed datasets.

Datasets	Training samples (Testing samples included)	Predicting samples	Input sequence length	Predicting steps
Dataset-1	CS2-36 CS2-37 CS2-38	CS2-35(0∼100%)	50	10
Dataset-2	CS2-36 CS2-37 CS2-38 CS2-35 (0∼25%)	CS2-35(25%∼100%)	50	10
Dataset-3	CS2-36 CS2-37 CS2-38 CS2-35 (0∼50%)	CS2-35(50%∼100%)	50	10
Dataset-4	CS2-36 CS2-37 CS2-38 CS2-35 (0∼75%)	CS2-35(75%∼100%)	50	10

the observed degradation points during time $T_{CS2-35} \cdot w/100 < t \leq T_{CS2-35}$. What is more, based on the training sample size proportion of each battery, 200 training samples are selected as testing samples to examine the network training accuracy. Because the input sequence length affects the recurrent times of network and the information of battery degradation, it is properly set as $L = 50$. What is more, in order to ensure the battery degradation security, the predicting steps is set as $k = 10$. Then, there is more preparation time for maintenance compared with one-step prognostics.

4.2. Results with different network types and dropout methods

In order to evaluate the Bi-LSTM network and dropout in depth dimension method, we build a set of networks with same hidden-cell number to predict the battery degradation. The training hyper-parameters are set as: learning rate $LR = 0.001$, batch size $BS = 200$, epoch times $ET = 1000$. Table 3 shows the different predicting results of different network architectures and dropout methods.

In Table 3, because testing samples are selected from training datasets, the testing MSEs of different networks and datasets are better than the corresponding predicting MSEs. With the same dropout method and dataset, the global predicting MSEs of different network architectures possess the following relationship: $\mathrm{MSE}_{RNN}^{dataset-i} > \mathrm{MSE}_{LSTM}^{dataset-i} > \mathrm{MSE}_{Bi-LSTM}^{dataset-i}$ $(i = 1, 2, 3, 4)$. What is more, with the same network architecture and dropout method, predicting MSEs possess the following relationship: $\mathrm{MSE}^{dataset-4} < \mathrm{MSE}^{dataset-3} < \mathrm{MSE}^{dataset-2} < \mathrm{MSE}^{dataset-1}$. Then, following conclusions could be obtained:

(a) LSTM cells possess "gate" structure, and LSTM nets could handle the sequence data better than RNNs which consist of standard recurrent cells, such as Sigma-cell and Tanh-cell.
(b) Bi-LSTM nets could make use of both forward and backward sequence information, through which more correlations of training samples could be learned, so Bi-LSTM nets could outperform LSTM networks.
(c) Different from common regression models and stochastic process models, network models are evolvable. For a specific Bi-LSTM architecture, when the network is trained

Table 3
The degradation predicting results of different nets and dropout methods.

Network	Dropout method	Dataset	Global testing MSE	Global predicting MSE
RNN (Sigma cell) 512–512–512	Dropout in depth dimension(50%)	Dataset-1	3.58×10^{-4}	3.88×10^{-4}
		Dataset-2	3.55×10^{-4}	3.78×10^{-4}
		Dataset-3	3.63×10^{-4}	3.71×10^{-4}
		Dataset-4	3.51×10^{-4}	3.54×10^{-4}
LSTM 512–512–512	Dropout in depth dimension(50%)	Dataset-1	3.18×10^{-4}	3.61×10^{-4}
		Dataset-2	3.21×10^{-4}	3.47×10^{-4}
		Dataset-3	3.19×10^{-4}	3.39×10^{-4}
		Dataset-4	3.18×10^{-4}	3.31×10^{-4}
Bi–LSTM 256–256–256	Dropout in depth dimension(50%)	Dataset-1	2.61×10^{-4}	3.02×10^{-4}
		Dataset-2	2.63×10^{-4}	2.84×10^{-4}
		Dataset-3	2.57×10^{-4}	2.76×10^{-4}
		Dataset-4	2.60×10^{-4}	2.65×10^{-4}

Table 4
Candidate model parameters setting.

Model	Network architecture	Batch size	Learning rate	Epoch times
M_0	256–256–256	200	0.001	500
M_1	256–256–256	100	0.001	500
M_2	256–256–256	300	0.001	500
M_3	256–256–256	200	0.0005	500
M_4	256–256–256	200	0.0015	500
M_5	256–256–256	200	0.001	400
M_6	256–256–256	200	0.001	600
M_7	384–384	200	0.001	500
M_8	192–192–192–192	200	0.001	500

Table 5
Training results of candidate models for different datasets.

Datasets	Global testing MSE and Prior probability (MSE/Prob.)			
	Dataset-1	Dataset-2	Dataset-3	Dataset-4
M_0	2.61×10^{-4}/11.00%	2.63×10^{-4}/11.15%	2.57×10^{-4}/10.68%	2.60×10^{-4}/13.80%
M_1	2.54×10^{-4}/11.23%	2.55×10^{-4}/10.16%	2.45×10^{-4}/13.43%	2.65×10^{-4}/10.53%
M_2	2.57×10^{-4}/11.68%	2.51×10^{-4}/11.45%	2.51×10^{-4}/11.06%	2.48×10^{-4}/10.11%
M_3	2.72×10^{-4}/11.74%	2.49×10^{-4}/11.09%	2.66×10^{-4}/9.09%	2.59×10^{-4}/11.32%
M_4	2.59×10^{-4}/12.07%	2.57×10^{-4}/13.05%	2.51×10^{-4}/11.19%	2.67×10^{-4}/10.26%
M_5	2.53×10^{-4}/10.50%	2.56×10^{-4}/10.14%	2.59×10^{-4}/10.49%	2.70×10^{-4}/10.34%
M_6	2.66×10^{-4}/11.16%	2.65×10^{-4}/11.80%	2.47×10^{-4}/11.45%	2.61×10^{-4}/10.75%
M_7	2.59×10^{-4}/10.12%	2.69×10^{-4}/10.78%	2.62×10^{-4}/11.87%	2.53×10^{-4}/10.72%
M_8	2.71×10^{-4}/10.50%	2.67×10^{-4}/10.38%	2.54×10^{-4}/10.74%	2.58×10^{-4}/12.26%

with newly observed degradation data, the prediction results could become more and more accurate.

4.3. Results with network model averaging method

In this sub-section, analyses of battery degradation trajectories predicting and RUL estimating are conducted to validate the proposed network model averaging method. Based on Tensorflow programming environment, Python 3.7 is adopted to program. The network training and degradation predicting is implemented on a desktop workstation equipped with two Intel Xeon Gold 5118 processors(16.5 M Cache, 2.30 GHz), and three graphic cards of NVIDIA TITAN RTX.

4.3.1. Network parameters setting and training results analyzing

Considering that model uncertainty could be caused by batch size, learning rate, epoch times and network framework, Table 4 gives settings of network training conditions.

Combining the above settings, we adopt BPTT as the training method, L2 error as the objective optimization function, and 50% dropout in depth dimension method as the regularization method in network training process. Training results of candidate models for different datasets are shown in Table 5.

In Table 5, the maximal testing MSE among the candidate networks for different datasets are $\mathrm{MSE}_{M_3}^{dataset-1}$, $\mathrm{MSE}_{M_7}^{dataset-2}$, $\mathrm{MSE}_{M_3}^{dataset-3}$ and $\mathrm{MSE}_{M_5}^{dataset-4}$, respectively. For battery CS2-35, the degradation path prediction MSE of exponential model adopted in [50–52] is 4.28×10^{-4}, and MSE of Wiener process used in [53] is 5.73×10^{-4}. Therefore, the training results of the candidate networks are valid. This means that the candidate network's parameters settings are reasonable. What is more, the maximal testing MSE errors of different datasets are $Error^{dataset-1} = \mathrm{MSE}_{M_3}^{dataset-1} - \mathrm{MSE}_{M_5}^{dataset-1} = 1.9 \times 10^{-5}$, $Error^{dataset-2} = \mathrm{MSE}_{M_7}^{dataset-2} - \mathrm{MSE}_{M_3}^{dataset-2} = 2.0 \times 10^{-5}$, $Error^{dataset-3} = \mathrm{MSE}_{M_3}^{dataset-3} - \mathrm{MSE}_{M_1}^{dataset-3} = 2.1 \times 10^{-5}$ and $Error^{dataset-4} = \mathrm{MSE}_{M_5}^{dataset-4} - \mathrm{MSE}_{M_2}^{dataset-4} = 2.2 \times 10^{-5}$, respectively. Because these testing MSE errors satisfy formula $Error^{dataset-i} \le 10\% \cdot \mathrm{MSE}_{minimum}^{dataset-i} (i = 1, 2, 3, 4)$, the trained candidate networks possess the approximative testing accuracy, and all of these obtained network models could be adopted in the following RUL predicting. Moreover, the corresponding prior probability of each candidate network is calculated through Eq. (8), and the obtained prior probability will be adopted to calculate the network probability in Eq. (11).

Table 6
The global predicting MSEs of different candidate networks.

Data sets	Global predicting MSE			
	Dataset-1	Dataset-2	Dataset-3	Dataset-4
M_0	3.02×10^{-4}	2.84×10^{-4}	2.76×10^{-4}	2.65×10^{-4}
M_1	3.11×10^{-4}	2.88×10^{-4}	2.71×10^{-4}	2.59×10^{-4}
M_2	3.03×10^{-4}	2.90×10^{-4}	2.80×10^{-4}	2.43×10^{-4}
M_3	2.97×10^{-4}	2.81×10^{-4}	2.77×10^{-4}	2.51×10^{-4}
M_4	3.07×10^{-4}	2.79×10^{-4}	2.72×10^{-4}	2.62×10^{-4}
M_5	2.91×10^{-4}	2.83×10^{-4}	2.71×10^{-4}	2.48×10^{-4}
M_6	3.04×10^{-4}	2.87×10^{-4}	2.69×10^{-4}	2.67×10^{-4}
M_7	2.95×10^{-4}	2.75×10^{-4}	2.70×10^{-4}	2.70×10^{-4}
M_8	2.91×10^{-4}	2.88×10^{-4}	2.77×10^{-4}	2.55×10^{-4}
Arithmetic averaging	2.88×10^{-4}	2.73×10^{-4}	2.62×10^{-4}	2.45×10^{-4}
Proposed model	2.81×10^{-4}	2.64×10^{-4}	2.55×10^{-4}	2.39×10^{-4}

4.3.2. Degradation trajectory predicting with network model averaging method

Based on the candidate network models and the prior probabilities, we could obtain the corresponding degradation prediction. The results of global predicting MSEs for the whole predicting datasets are shown in Table 6.

In order to show the advantages of our network model averaging method, we further adopt the arithmetic averaging method as a comparison. Table 6 shows that model $M_0 \sim M_8$ possess the similar predicting accuracy for the same dataset, and the proposed model averaging method has higher accuracy. For different datasets, predicting MSEs generally possess the relationship: $\mathrm{MSE}^{dataset-4} < \mathrm{MSE}^{dataset-3} < \mathrm{MSE}^{dataset-2} < \mathrm{MSE}^{dataset-1}$. Therefore, following conclusions could be obtained:

(1) Both arithmetic averaging method and the proposed model averaging method could reduce the random errors of the predicted degradation values. They have higher global predicting accuracy compared with single network model.
(2) Our predicting step is $k = 10$, the predicting results are affected by the training accuracy to some extent. Therefore, compared with the arithmetic averaging method which does not consider testing accuracy, our method shows better properties in dealing with model uncertainty and could obtain the higher predicting accuracy.
(3) Different training datasets possess different quantities of CS2-35 degradation samples, and the predicting accuracy of dataset that contains more battery CS2-35 degradation samples is higher, when predicting the degradation trajectory of CS2-35. This means that the network models are evolvable with newly observed degradation data.

4.3.3. RUL predicting with network model averaging method

Based on the above predicted degradation values, we adopts the algorithm in Fig. 9 to predict the RUL of battery CS2-35. In the RUL predicting, we select 4 starting points to show the advantages of our model averaging method. Let $SP_{i,j}$ denote the jth starting point of Dataset-i. Then, considering the length of battery CS2-35 degradation trajectory, we set $SP_{i,1} = 100$, $SP_{i,2} = 300$, $SP_{i,3} = 500$, and $SP_{i,4} = 700$. What is more, in order to utilize the degradation data effectively and sufficiently, we set failure threshold as $Threshold = 0.5$, then the corresponding failure time is $t = 802$. Table 7 shows the predicted RUL results of different datasets and network models, and the optimal RUL results of different datasets and network models at each RUL predicting starting point are marked in red.

In Table 7, the network models perform well in RUL predicting, the obtained RULs satisfy $|RUL_{Prediction} - RUL_{Real}| < RUL_{Real}$, where $RUL_{Prediction}$ is the predicted RUL and RUL_{Real} is the real RUL. Then, in order to evaluate the results, we define RUL predicting accuracy as:

$$RUL_{Accuracy} = 1 - \sqrt{(RUL_{Prediction} - RUL_{Real})^2}/RUL_{Real} \tag{14}$$

where a higher $RUL_{Accuracy}$ means a more accurate predicting result. Table 7 shows that model $M_0 \sim M_8$ possess the approximate RUL predicting accuracy for the same dataset at each starting point $SP_{i,j}$ $(1 \le i \le j \le 4)$, and the proposed model averaging method has higher predicting accuracy. For different RUL predicting starting points of each dataset, the predicting RUL errors follow the relationship: $Error^{SP_{i,m}} > Error^{SP_{i,n}}$ $(1 \le m < n \le 4)$. What is more, for different dataset, predicting RUL errors generally possess the relationship: $Error_{SP_{k,j}}^{dataset-k} > Error_{SP_{l,j}}^{dataset-l}$ $(1 \le k < l \le 4)$. Therefore, following conclusions could be obtained:

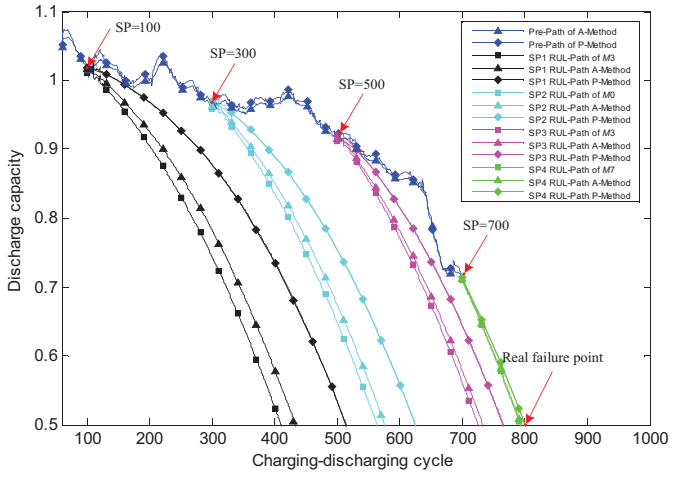
(1) Both arithmetic averaging method and the proposed method could reduce the predicting RUL errors. They have more precise predicting RUL compared with single network model.
(2) The predicting results are affected by the training accuracy, therefore, the proposed averaging method which considers testing accuracy shows higher RUL predicting accuracy compared with the arithmetic averaging method.
(3) When the RUL predicting starting points are closer to the failure time, the degradation characteristics are more similar and the accuracy of RUL predicting would be higher.
(4) When the training dataset contains more degradation information of target system, the network model could evolve better and higher RUL predicting accuracy could be obtained.

In order to display the RUL predicting results distinctly, the optimal RUL predicting results among these single network models are selected the one comparing object at each RUL starting point. Then, the RUL predicting results of the proposed model, arithmetic averaging method and the selected optimal network are shown in Fig. 11.

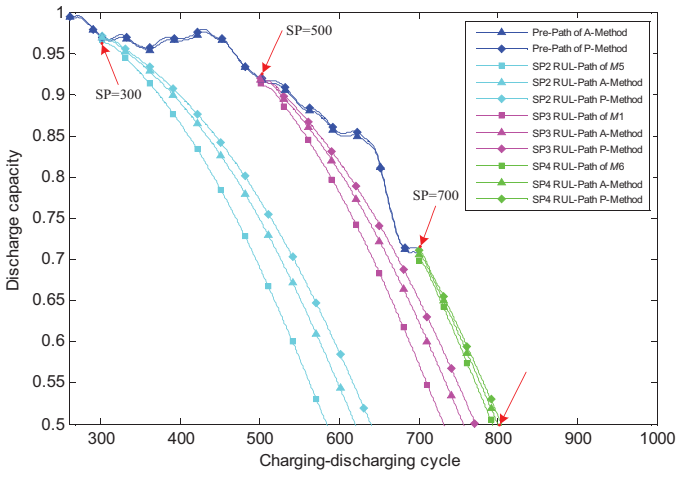
In Fig. 11, the A-Method and P-Method denote the arithmetic averaging method and the proposed averaging method, respectively. The Pre-Path denotes the predicted degradation values with real observed degradation data, and the RUL-Path is the obtained trajectory with local predicted degradation values when computing the RUL at starting point $SP_{i,j}$. Because the network input sequence length is $L = 50$ and the predicting step is $k = 10$, the starting point of the predicting trajectories are $t = 60$ for Dataset-1. What is more, for calculating simplicity, degradation samples of battery CS2-35 at time $t \in [1, 260]$, $t \in [1, 460]$ and $t \in [1, 660]$ are included in Dataset-2, Dataset-3, and Dataset-4, respectively. Therefore, the starting point of predicting trajectories in Fig. 11(b)–11(d) are $t = 261$, $t = 461$ and $t = 661$, respectively.

Table 7
The obtained RUL(error) and predicting accuracy.

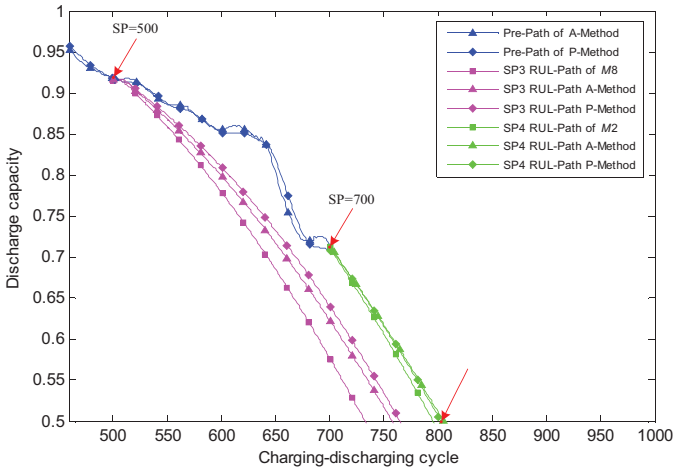
Datasets	Dataset-1					Dataset-2				Dataset-3			Dataset-4
	$SP_{1.1}$	$SP_{1.2}$	$SP_{1.3}$	$SP_{1.4}$	Mean Accuracy	$SP_{2.2}$	$SP_{2.3}$	$SP_{2.4}$	Mean Accuracy	$SP_{3.3}$	$SP_{3.4}$	Mean Accuracy	$SP_{4.4}$
Real RUL	702	502	302	102		502	302	102		302	102		102
M_0	242(−460); 0.3447	263(−239); 0.5239	172(−130); 0.5695	81(−21); 0.7941	0.5212	262(−240); 0.5219	368(+66); 0.7815	93(−9); 0.9118	0.7384	221(−81); 0.7318	114(+12); 0.8824	0.8071	96(−6); 0.9412
M_1	249(−453); 0.3547	182(−320); 0.3625	178(−124); 0.5894	83(−19); 0.8137	0.5301	276(−226); 0.5498	231(−71); 0.7649	92(−10); 0.9020	0.7223	376(+74); 0.7550	87(−15); 0.8529	0.8040	109(+7); 0.9314
M_2	238(−464); 0.3390	772(+270); 0.4622	381(+79); 0.7384	115(+13); 0.8725	0.6030	249(−253); 0.4960	228(−74); 0.7550	117(+15); 0.8529	0.7013	225(−77); 0.7450	95(−7); 0.9314	0.8515	90(−12); 0.8824
M_3	310(−392); 0.4416	189(−313); 0.3765	225(−77); 0.7450	116(+14); 0.8627	0.6531	283(−219); 0.5637	211 (−91); 0.6987	87(−15); 0.8529	0.7051	212(−90); 0.7020	91(−11); 0.8922	0.7971	93(−9); 0.9118
M_4	1097(+395); 0.4373	218(−284); 0.4343	218(−84); 0.7219	112(+10); 0.9020	0.6239	732(+230); 0.5418	216(−86); 0.7152	115(+13); 0.8725	0.7475	230(−72); 0.7616	110(+8); 0.9216	0.8416	112(+10); 0.9020
M_5	205(−497); 0.2920	209(−293); 0.4163	157(−145); 0.5199	85(−17); 0.8333	0.5154	284(−218); 0.5657	232(−70); 0.7682	113(+11); 0.8922	0.7341	234(−68); 0.7748	110(+8); 0.9216	0.8482	97(−5); 0.9510
M_6	226(−476); 0.3219	221(−281); 0.4402	388(+86); 0.7152	83(−19); 0.8137	0.5728	286(−216); 0.5697	224(−78); 0.7417	93(−9); 0.9118	0.7280	219(−83); 0.7252	112(+10); 0.9020	0.8136	106(+4); 0.9608
M_7	209(−493); 0.2977	234(−268); 0.4661	175(−127); 0.5795	92(−10); 0.9020	0.5515	265(−237); 0.5279	369(+67); 0.7781	90(−12); 0.8824	0.7295	372(+70); 0.7682	104(+12); 0.8824	0.8253	110(+8); 0.9216
M_8	221(−481); 0.3148	196(−306); 0.3904	203(−99); 0.6722	79(−23); 0.7745	0.5380	243(−259); 0.4841	225(−77); 0.7450	91(−11); 0.8922	0.7071	233(−69); 0.7715	112(+10); 0.9020	0.8235	96(−6); 0.9412
Arithmetic averaging	333(−369); 0.4744	276(−226); 0.5498	233(−69); 0.7715	94(−8); 0.9216	0.6793	320(−182); 0.6375	256(−46); 0.8477	99(−3); 0.9706	0.8186	258(−44); 0.8543	105(+3); 0.9706	0.9125	101(−1); 0.9902
Proposed model	414(−288); 0.5897	325(−177); 0.6474	265(−37); 0.8775	100(−2); 0.9804	0.7738	339(−163); 0.6753	271(−31); 0.8974	104(+2); 0.9804	0.8510	265(−37); 0.8775	103(+1); 0.9902	0.9339	102(0); 1.0000



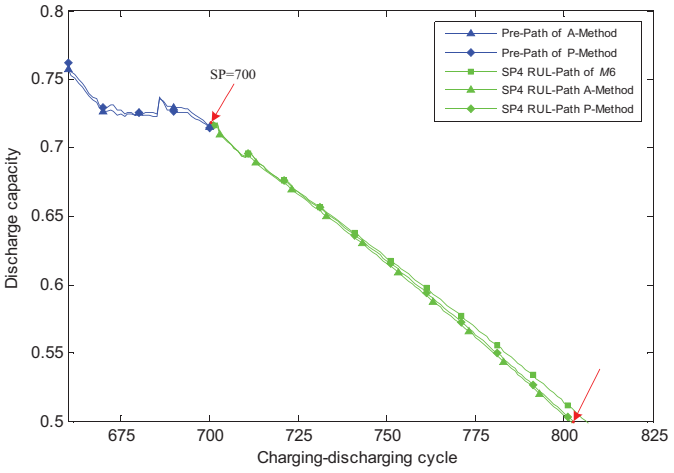
(a) The results of Dataset-1



(b) The results of Dataset-2



(c) The results of Dataset-3



(d) The results of Dataset-4

Fig. 11. The obtained RULpredicting path and the corresponding predicted degradation trajectory, (a) The results of Dataset-1, (b) The results of Dataset-2, (c) The results of Dataset-3, (d) The results of Dataset-4.

5. Conclution

In this paper, we propose a novel network model averaging method to deal with the network model uncertainty in RUL prediction of non-life-cycle labeled dataset. The proposed network model averaging method adopts the training accuracy and degradation predicting accuracy as the model probability updating parameters when dealing with network model uncertainty, because the RUL predicting algorithm is directly influenced by the predicted degradation values. The experiments were conducted on datasets from the CALCE Battery Research Group. Bi-LSTM networks that could learn the temporal correlation informations in both forward and backward time directions simultaneously were constructed to verify the effectiveness of proposed network model averaging method. In the training process, L2 regularization and dropout in depth dimension method are used to prevent network overfitting. As for the network constructing and training, the results show that Bi-LSTM network outperforms RNN and LSTM network with the same hidden layers and cells, and dropout in depth dimension method could improve the testing and predicting accuracy. In predicting the degradation trajectory and RUL, the proposed network model averaging method could obtained more precise results compared with single network model. This paper focuses on the probability updating of candidate models based on networks' outputs. In future work, we may pay attention to update the inner weights of network and inner parameters of basic cells.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yong Yu: Investigation, Methodology, Software, Visualization, Validation, Writing - original draft. **Changhua Hu:** Project administration, Supervision. **Xiaosheng Si:** Conceptualization,

Investigation, Writing - review & editing, Supervision. **Jianfei Zheng:** Resources, Writing - review & editing. **Jianxun Zhang:** Visualization, Writing - review & editing.

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